

TRANSPORT MANAGEMENT SYSTEM (TMS)

PROJECT PROPOSAL DOCUMENT – PREMIUM VERSION

1. PROPOSAL SUMMARY

We propose to develop a **Premium Enterprise Transport Management System (TMS)** — a complete digital logistics and fleet ecosystem with mobile apps (Customer + Driver + Operations), AI-powered dispatch and route optimization, IoT vehicle diagnostics, FASTag integration, full freight ERP, and enterprise-grade analytics. Designed for **large logistics companies, fleet enterprises, and multi-depot transport operators (300+ vehicles)** requiring maximum scalability, automation, and performance.

Ideal For: Large logistics companies, fleet enterprises, last-mile delivery platforms, multi-depot transport operators (300+ vehicles)

2. OBJECTIVES

- Build an enterprise-grade, fully automated transport and logistics platform
- Achieve complete real-time fleet visibility across all depots and routes
- Implement AI-powered dispatch, route optimization, and demand forecasting
- Automate compliance, billing, freight operations, and driver management
- Deliver deep business intelligence for strategic fleet decisions

3. SCOPE OF WORK

3.1 Website + Customer & Corporate Portal (Enterprise)

- Premium transport and logistics website
- Customer self-service booking and tracking portal
- Corporate client dashboard for bulk booking
- Vendor / partner onboarding portal
- Live freight and shipment tracking page
- Service rate calculator for customers

Cost: Rs. 85,000 - Rs. 1,35,000

3.2 Real-Time GPS + IoT Vehicle Telematics

- Real-time GPS live tracking for entire fleet
- OBD-II device integration for engine diagnostics
- Speed, fuel level, temperature, and load sensors
- Geo-fence alerts and unauthorized movement detection

- Historical route playback and trip analysis
- Driver behaviour scoring (harsh braking, acceleration)

Cost: Rs. 1,20,000 - Rs. 2,00,000

3.3 AI Dispatch, Route Optimization & Planning

- AI-powered smart vehicle-to-trip auto dispatch
- Multi-stop and multi-vehicle route optimization
- Traffic-aware and weather-aware dynamic re-routing
- Load and capacity optimization for freight
- AI-based fleet demand forecasting and planning
- SLA breach prediction and proactive alerts

Cost: Rs. 1,10,000 - Rs. 1,80,000

3.4 Enterprise Fleet & Asset Management

- Complete vehicle lifecycle management
- Predictive maintenance with IoT-based triggers
- Multi-depot fleet management and inter-depot transfer
- Asset depreciation and residual value tracking
- FASTag auto-deduction and toll reconciliation
- Insurance, permit and fitness management with auto-renewal

Cost: Rs. 90,000 - Rs. 1,50,000

3.5 Driver Management + HRMS

- Full driver lifecycle management
- HRMS with payroll, incentive and deduction automation
- Behaviour-based performance scoring and rewards
- Trip allowance, advance and settlement management
- Driver training and certification tracking
- Medical fitness and license renewal management

Cost: Rs. 80,000 - Rs. 1,30,000

3.6 Freight & Supply Chain Management (ERP)

- End-to-end consignment lifecycle management
- LR generation, E-way bill and invoice automation
- Digital POD (Proof of Delivery) with signature
- Multi-leg and intermodal freight tracking
- Vendor, agent and broker management
- Freight rate contract and bid management

Cost: Rs. 1,00,000 - Rs. 1,60,000

3.7 Billing, Finance & ERP Integration

- Auto GST-compliant invoice and bill generation
- Corporate account credit and billing management
- Tally / Zoho Books / SAP integration
- Fuel card reconciliation and expense management
- P&L; per vehicle, route, depot and division
- Financial dashboards with forecasting

Cost: Rs. 90,000 - Rs. 1,50,000

3.8 Mobile Applications (Android / iOS)

- Customer App: Booking, live tracking, invoice download, support
- Driver App: Trip acceptance, navigation, POD upload, earnings, attendance
- Operations App: Dispatch, fleet view, alerts, reports, driver communication

Cost: Rs. 2,50,000 - Rs. 4,50,000

4. ADDITIONAL FEATURES

4.1 AI Analytics & Business Intelligence

- Fleet utilization and profitability heatmaps
- Route-wise and customer-wise revenue analytics
- Predictive breakdown and maintenance cost alerts
- Driver safety and risk scoring

Cost: Rs. 90,000 - Rs. 1,40,000

4.2 FASTag, E-Way Bill & Compliance Suite

- FASTag real-time toll deduction and balance alerts
- Automated E-way bill generation and extension
- VAHAN API for vehicle verification
- RTO compliance checklist and alerts

Cost: Rs. 55,000 - Rs. 85,000

4.3 WhatsApp + SMS + Email + Push Notifications

- Full customer and driver communication automation
- Live tracking link and ETA updates via WhatsApp
- Delivery and payment confirmation alerts
- Fleet and compliance alert dashboard

Cost: Rs. 40,000 - Rs. 65,000

4.4 Security, Audit & Compliance

- Multi-level role-based access with approvals
- Full trip, financial and system audit trail

- AES-256 data encryption
- Motor Vehicle Act and CMVR compliance tools

Cost: Rs. 50,000 - Rs. 85,000

4.5 Customer & Partner API Integration

- Open API for corporate client ERP integration
- Third-party logistics (3PL) portal integration
- Webhook support for real-time event notifications

Cost: Rs. 45,000 - Rs. 75,000

5. TECHNOLOGY STACK (MERN + AI + MOBILE)

- **Frontend:** React.js + Next.js (SSR)
- **Backend:** Node.js + Express.js (Microservices)
- **Database:** MongoDB + Redis + TimescaleDB (GPS data)
- **Mobile Apps:** React Native / Flutter
- **GPS / Telematics:** Google Maps + HERE Maps + OBD API
- **AI/ML:** Route optimization + demand forecasting + driver behaviour models
- **Compliance:** NIC E-way Bill API + VAHAN API + FASTag API
- **Hosting:** AWS (EC2, S3, CloudFront, RDS)

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 10,00,000 - Rs. 18,00,000+ (One-time)
Infrastructure	AWS Hosting: Rs. 15,000 - Rs. 60,000/month
Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	GPS & Telematics API: Rs. 5,000 - Rs. 20,000/month
Infrastructure	CDN & Backup: Rs. 18,000/year
Maintenance Cost	Rs. 60,000 - Rs. 1,20,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	2 Weeks
Development	14-18 Weeks
Testing	3-4 Weeks

Deployment	1-2 Weeks
Total Duration	4-6 Months

8. LIMITATIONS

- High initial investment
- Requires dedicated IT and operations support team
- GPS hardware device cost is additional (per vehicle)
- Third-party API and infrastructure costs are recurring
- Comprehensive staff onboarding and training mandatory

9. FUTURE UPGRADES

- Blockchain-based freight and supply chain traceability
- Drone delivery management integration
- AI-based carbon footprint tracking and green fleet reporting
- Hyperlocal last-mile delivery optimization
- EV fleet charging station management integration

10. SUPPORT AND WARRANTY

- 3 months free support
- Priority bug fixing
- Dedicated account manager and support team
- On-site deployment and training (optional)

11. TERMS AND CONDITIONS

- 40% advance payment
- 30% mid-phase
- 30% after completion
- Additional modules charged separately

12. CONCLUSION

This Premium TMS is a **complete enterprise transport and logistics platform** — combining real-time GPS telematics, AI-powered dispatch and route optimization, IoT vehicle diagnostics, freight ERP, FASTag and compliance tools, mobile apps for customers/drivers/operations, and deep business intelligence, built to power India's next-generation transport and logistics enterprises at scale.

13. CONTACT DETAILS

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